source ..... instruo manual v2
rev 2026.10

lúbadh

_travelers guide _for navigating time loops

2	10	4
decks · stereo link	min per loop, per deck	taps per deck

`[ rec ]` press `[ rec ]` hold `[ shift ]` = retrig/shift opp. = the other deck's button › then
◆ = preset / mode only

every control is per deck: left and right are identical

combos shown for one deck · mirror them for the other

amber = hold

01 transport — lbh_01

`[ rec ]`	record › press again to stop & play. press while playing = **overdub**
`[ erase ]`	erase this deck's loop
`[ shift ]`	jump back to start
turn `[ speed ]`	speed & direction: center = stall, cw = forward to 4×, ccw = reverse to 4×
`[ shift ]` + turn `[ speed ]`	knob is muted until release, then jumps straight to the new speed
`[ link ]` up	right deck mirrors left · saves and loads stereo
`[ shift ]`	adds a playhead at start (up to 4)
◆ multi-tap presets	

02 recording — lbh_02

`[ erase ]` + `[ rec ]`	**punch-in**: replaces audio while recording
`[ shift ]` `[ shift ]` + `[ rec ]`	**input monitoring**: on (rec lit) › armed (rec pulses) › off
armed `[ rec ]`	arm › record › stop & play · `[ erase ]` disarms, loop kept
`[ shift ]` + opp. `[ rec ]`	looping ← → **one-shot record** (records one pass, then stops)
`[ rec ]` jack	trig starts / stops recording, like `[ rec ]` (latching)
`[ rec ]` jack	gated: records only while the gate is high
◆ rec jack = gated	

* overdub keeps the old layer at 0.9 by default, so layers fade a little each pass (dub level, 04).

__fig.01 : speed

white speed LEDs = notches



4× rev	½ rev	stall	½	1×	2×	4×
+24 st	-12 st	–	-12 st	0 st	+12 st	+24 st

speed sets pitch: every doubling = +1 octave · reverse plays backwards

03 playback — lbh_03

`[ shift ]` + opp. `[ erase ]`	looping ← → **one-shot playback** (rec blinks white / mixed)
`[ shift ]` or `[ retrig ]` jack	one-shot: play one pass of start › length
touch the reel	warble or stall the tape · trimmer sets sensitivity
`[ speed ]` cv	±5 v, summed with the knob
`[ erase ]` + opp. `[ erase ]`	flip playback direction · speed LEDs blink
◆ v/oct speed mode only	
`[ speed ]` cv	tracks 1v/oct: sequence the loop's pitch
◆ v/oct · seq presets	

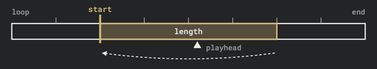
04 time knob — lbh_04

`[ shift ]` + `[ rec ]`	cycle the time function (amber LED by `[ clk out ]` shows which)
LED off	**clock divisions**: triggers per loop at `[ clk out ]`: 1–12, 16, 24, 32, 64
LED solid	**dub level**: how much old audio survives each overdub. high = delay feedback
LED pulsing	**user parameter**: crossfade, tape or slew (preset picks)



05 start · length — lbh_05

turn `[ start ]`	where the loop begins inside the recording
turn `[ length ]`	how much plays before jumping back. tiny = wavetable-like tones
`[ shift ]` + turn `[ time ]`	**quantise** start & length: lit LED sections = 1/2, 1/4, 1/8 …
`[ shift ]` + turn `[ start ]`	fine move that ignores the quantise grid (same for `[ length ]`)



ticks = quantise grid (1/8)

06 save · load — lbh_06

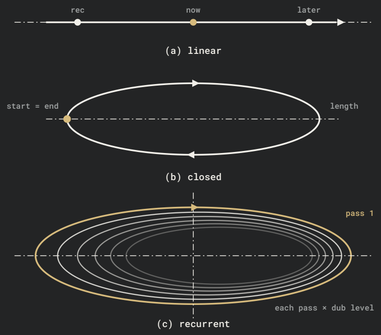
`[ shift ]` `[ erase ]` + `[ rec ]`	**save** to usb › saved (folder 0, newest first)
`[ shift ]` + `[ erase ]`	**load menu**
in the menu	own `[ time ]` = loop · opposite `[ time ]` = folder · `[ rec ]` loads · hold `[ shift ]` previews 5 s · `[ erase ]` exits
`[ link ]` up	saves and loads stereo (mono loads of stereo are summed)
power-on	last session reloads automatically

07 presets — lbh_07 · v2.1

`[ shift ]` `[ shift ]` + `[ erase ]`	**preset menu** for that deck
turn `[ time ]`	pick 1 of 12 (shown on the LED ring) · `[ rec ]` loads · `[ erase ]` cancels
usb › presets	.txt files, sorted 01–12. build your own with the preset editor

__fig.02 : the return

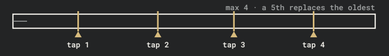
mm_02



08 multi-tap — lbh_08

◆ multi-tap presets only

`[ shift ]`	new playhead from start
`[ erase ]` + `[ shift ]`	delete the taps
`[ erase ]` trig, cv exp.	removes taps remotely



09 sync · clock — lbh_09

`[ clk out ]`	triggers per loop (set by clock div)
`[ retrig ]` jack	resets the playhead to start: sync to an external clock
`[ rec ]` jack	external clock sets the loop length
◆ clockable preset	
`[ erase ]` + `[ rec ]`	tap tempo instead of punch-in
◆ erase + rec = tap tempo	

10 aux · routing — lbh_10

aux in fader	sends `[ aux ]` in to the left ← → right deck input (+ `[ cv ]` ±5 v)
aux out fader	blends left ← → right outputs at `[ aux ]` out (+ `[ cv ]` ±5 v)
rear switch	aux out fader pre or post the output level, per deck



11 reset · calibrate — lbh_11

`[ shift ]` `[ shift ]` + `[ rec ]` `[ rec ]`	**reset modes** to default · audio kept · speed LEDs pulse
power on, hold `[ shift ]`×2 `[ erase ]`×2	**calibration**: speed knobs min, 0 v › `[ rec ]` each deck, then 1 v and 3 v; knobs max, no cv › `[ rec ]` = 5 v
amber blinks	voltage out of range: defaults used

pdf print

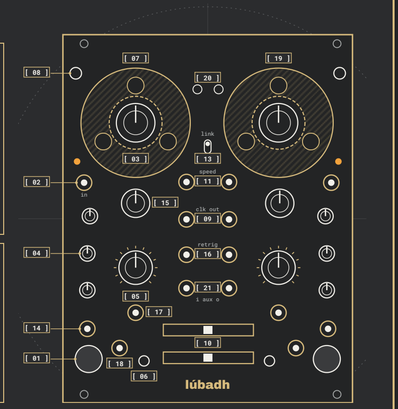
__fig.03 : front panel

01–11 = sections · 13–21 = hardware

left deck labelled · right deck mirrors it

`[ 01 ]`	**_rec :** record · overdub · stop
`[ 02 ]`	**_in :** input + level
`[ 03 ]`	**_speed :** rate & direction
`[ 04 ]`	**_time :** clock div · dub · user
`[ 05 ]`	**_start :** loop start
`[ 06 ]`	**_erase :** erase · save / load
`[ 07 ]`	**_LED ring :** position · presets
`[ 08 ]`	**_retrig/shift :** restart · shift
`[ 09 ]`	**_clk out :** clock triggers
`[ 10 ]`	**_aux faders :** in / out crossfade
`[ 11 ]`	**_speed cv :** ±5 v · 1v/oct

`[ 13 ]`	**_link :** stereo mirror
`[ 14 ]`	**_out :** output + level
`[ 15 ]`	**_length :** loop length
`[ 16 ]`	**_retrig :** jack, restart
`[ 17 ]`	**_cv :** aux fader cv
`[ 18 ]`	**_rec :** jack, record gate
`[ 19 ]`	**_reel :** capacitive touch
`[ 20 ]`	**_speed LEDs :** notches
`[ 21 ]`	**_aux :** in · out jacks



_where things hide

12 · lbh_12

preset files

usb › presets › .txt · instruo preset editor

- **speed** record variable · fixed, play follow · hold
- **speed control** notched · stepped · smooth · v/oct, markers 0–4
- **multi-tap** engine off · on
- **time user** crossfade · speed slew · tape emulation
- **rec jack** latching · gated · external clock
- **ext clock** average, resolution, timeout, divisions
- **quantise** all · even · odd · powers of two
- **erase + rec** punch-in · tap tempo
- **timing** min loop, crossfade, retrig delay, max dub
- **tape** wow / flutter, crinkle, age, hysteresis, wear, saturation, reverb
- **input filter** low cut freq + q, high cut
- **touch** off · stall · dip

usb disk

- **samples** 12 folders × 12 loops
- **saved** folder 0, newest first
- **presets** first 12 show in the menu

tails & tags

250 ms, in the file name

- **{s}** silent tail (default): keeps timing
- **{n}** no tail: smoothest, timing loosens
- **{c}** copy tail: keeps timing, may click

hardware

- **cv expander** `[ length ]` `[ start ]` `[ time ]` cv + `[ erase ]` trig, per deck
- **rear pcb** touch on / off · pre / post output level
- **front** touch sensitivity trimmer

factory presets

ring = what the LED ring shows in the preset menu

- **01 tape looper** · tape feel · varispeed rec
- **02 clean** · crossfade on time · little tape
- **03 classic** · v1 behaviour · fixed rec speed
- **04 multi-tape** · multi-tap delay lines
- **05 clockable** · sync to an external clock
- **06 octave delay** · pitch-shifting delay line
- **07 seq mono** · 1v/oct · retrig delay
- **08 seq poly** · multi-tap 1v/oct · speed hold
- **09 melontron** · 4-note mellotron chords
- **10 broken tape** · maxed-out warble & grit